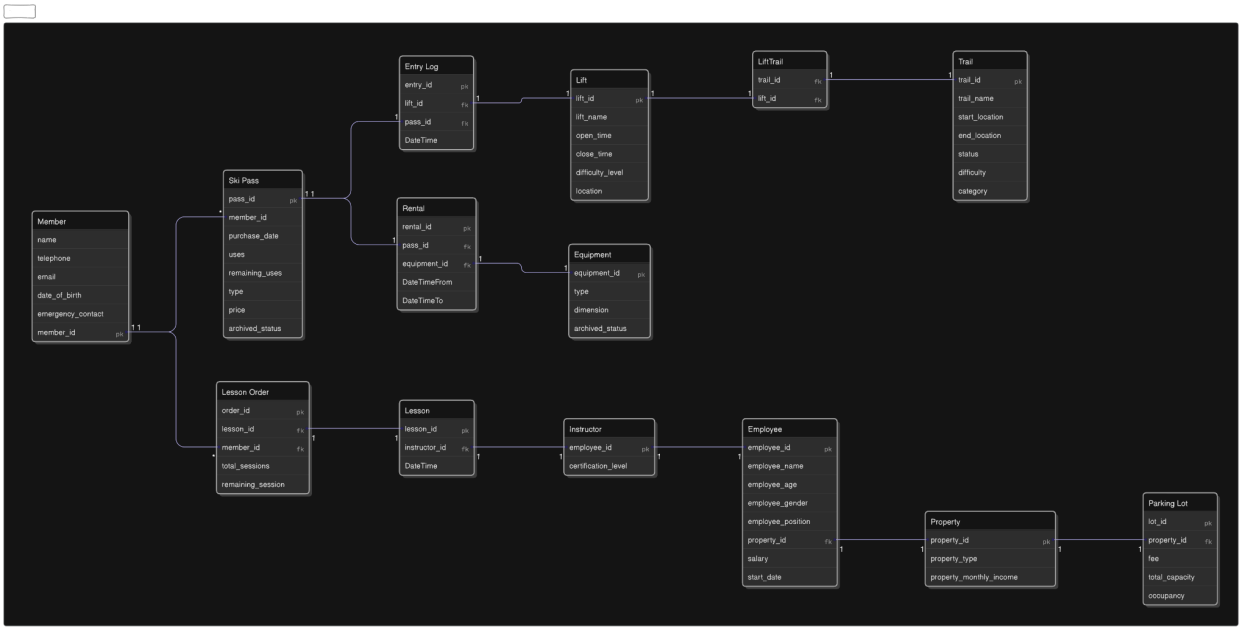
Conceptual Database Design



Logical Database Design

Property

property_id	property_type	property_monthly_income	location

ParkingLot

lot_id	property_id	fee	total_capacity	occupancy

Member

member_id	member_name	telephone	email	date_of_birth	emergency_contact

SkiPass

pass_id	member_id	purchase_date	total_uses	remaining_uses	pass_type	price	archived_status

Equipment

equipment_id	equipment_type	dimension	archived_status

Rental

rental_id	pass_id	equipment_id	datetime_from	datetime_to

Trail

trail_id	trail_name	start_location	end_location	status	difficulty	category

Lift

lift_id	lift_name	open_time	difficulty_level	close_time	location

LiftTrail

lift_id	trail_id

EntryLog

entry_id	lift_id	pass_id	datetime

Employee

employee_id	employee_name	employee_age	employee_gender	employee_position	salary	start_date

Instructor

employee_id	certification_level

Lesson

lesson_id	instructor_id	lesson_time

LessonOrder

order_id	lesson_id	member_id	total_sessions	remaining_sessions

Normalization Analysis

Property

FDs:

- property_id → property_type, property_monthly_income, location

3NF Justification: all attributes are solely functionally determined by the primary key.

ParkingLot
FDs:

- lot_id → property_id, fee, total_capacity, occupancy

3NF Justification: all attributes are solely functionally determined by the primary key.

Member
FDs:

- member_id → member_name, telephone, email, date_of_birth, emergency_contact

3NF Justification: all attributes are solely functionally determined by the primary key.

SkiPass
FDs:

- pass_id → member_id, purchase_date, total_uses, remaining_uses, pass_type, price, archived_status

3NF Justification: all attributes are solely functionally determined by the primary key.

Equipment
FDs:

- equipment_id → equipment_type, dimension, archived_status

3NF Justification: all attributes are solely functionally determined by the primary key.

Rental
FDs:

- rental_id → pass_id, equipment_id, datetime_from, datetime_to

3NF Justification: all attributes are solely functionally determined by the primary key.

Trail
FDs:

- trail_id → trail_name, start_location, end_location, status, difficulty, category

3NF Justification: all attributes are solely functionally determined by the primary key.

Lift
FDs:

- lift_id → lit_name, open_time, difficulty_level, close_time, location

3NF Justification: all attributes are solely functionally determined by the primary key.

LiftTrail
FDs: {trail_id, lift_id} → {trail_id, lift_id}

3NF Justification: The only functional dependency is a reflexive dependency of the primary key on itself, therefore it inherently satisfies 3NF.

EntryLog
FDs:

- entry_id → lift_id, pass_id, datetime

3NF Justification: all attributes are solely functionally determined by the primary key.

Employee
FDs:

- employee_id → employee_name, employee_age, employee_gender, employee_position, salary, start_date

3NF Justification: all attributes are solely functionally determined by the primary key.

Instructor
FDs:

- employee_id → certification_level

3NF Justification: all attributes are solely functionally determined by the primary key.

Lesson
FDs:

- lesson_id → instructor_id, lesson_time

3NF Justification: all attributes are solely functionally determined by the primary key.

LessonOrder
FDs:

- order_id → lesson_id, member_id, total_sessions, remaining_sessions

3NF Justification: all attributes are solely functionally determined by the primary key.

Custom Query

Our custom query is: “What is the wage expense of property X?”
This query takes the sum of the wages of all employees working at property X.
This query is useful for calculating a property’s operating costs.